

Question	Marking Guidance	Mark	Comments
03.1	1. Stirrer distributes <u>heat</u> (energy); 2. Insulation/space/air reduces loss/gain of <u>heat</u> OR Insulation/space/air reduces conduction/convection; 3. Water has high (specific) heat capacity;	2 max	1. Accept stirrer ensures equal/even temperature or prevents build up of 'hot spots'. 2. Reject vacuum.
03.2	3.28 / 3.3 (kJ g ⁻¹) = 2 marks ;; Incorrect answer but shows 328 / 33 (ignore any subsequent numbers and decimal point) = 1 mark OR Incorrect answer of 6.56 / 6.6 (kJ g ⁻¹) (ignore any subsequent numbers) = 1 mark ;	2	
03.3	1. (Light is) reflected; 2. (Light is) wrong wavelength; 3. (Light) misses chlorophyll/ chloroplasts/photosynthetic tissue; 4. CO ₂ <u>concentration</u> or temperature is a <u>limiting factor</u> .	2 max	1. Light is not absorbed on its own is not enough. 2. Accept frequency for wavelength. 2. Accept reference to absorbing specified wavelengths/frequencies.
03.4	1. ATP; 2. Reduced NADP;	2	Accept 1 and 2 in either order. 2. Reject Reduced NAD. 2. Accept NADPH/NADPH ₂ .
03.5	Correct answer of 1.31/1.3 × 10 ⁸ (ignoring any subsequent numbers after 1.31) = 2 marks ;; Incorrect answer but shows 2 ¹⁶ = 1 mark	2	

	OR 65536 in any correct numerical form = 1 mark OR Incorrect answer but shows 131 ignoring any subsequent numbers and ignoring any decimal point = 1 mark;		
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